Abstract

The disposal of lithium-based drone batteries presents a significant environmental challenge due to the presence of heavy metals and hazardous substances. Effective management strategies are essential to reduce pollution and mitigate operational risks associated with improper handling. This study proposes an optimal waste management strategy for Skywalker drone batteries using a Linear Programming (LP) approach. The model incorporates three waste management options: recycling, temporary storage, and final disposal. It also accounts for facility capacity limitations, environmental regulations, and cost constraints. The simulation results demonstrate that the LP model provides an optimal waste allocation scheme. Compared to conventional waste management methods, the LP-based strategy reduces environmental impact and achieves higher cost efficiency. The findings highlight the effectiveness of LP modeling as a decision-support tool for waste management planning. The study recommends the adoption of an LP-based integrated management framework to support future environmental and operational decisions in drone technology.

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